

01_univariate_model

Univariate approach for KM

Import Required Library

```
library(survival)
library(survminer)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: ggpubr
```

```
##
```

```
## Attaching package: 'survminer'
```

```
## The following object is masked from 'package:survival':
```

```
##
```

```
##      myeloma
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

```
library(mice)
```

```
## Registered S3 method overwritten by 'lme4':
```

```
##      method          from
```

```
##      na.action.merMod car
```

```
##
```

```
## Attaching package: 'mice'
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
##      filter
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      cbind, rbind
```

```
dataset_path <- "/Users/sorenbasnet/Documents/Github/STAT_443_FINAL_PROJECT/data/data_patient_file_cna_
```

```
data <- read.table(file.path(dataset_path, "patient_file_cna_rai.txt"), header=TRUE, sep=" ")
```

Missing Data

```
summary(as.factor(data$OS_STATUS))
```

```
## DECEASED    LIVING      NA's
##      286      722      140
```

```
summary(as.factor(data$IGHV_MUTATION_STATUS))
```

```
## mutated unmutated      NA's
##      581      543      24
```

```
data$death_5yr_status_num <- ifelse(data$OS_STATUS == "DECEASED" & data$OS_MONTHS <= 60, 1, 0)
data$os_months_5yr <- pmin(data$OS_MONTHS, 60)
```

IGHV survival

```
impute_data <- data[, c("IGHV_MUTATION_STATUS", "X11q_status", "X17p_status", "X18p_status", "X19q_status", "X20p_status", "X2p_status", "X4p_status", "X6q_status", "X8p_status", "tri12_status", "RAI_AT_DIAGNOSIS", "RAI_AT_SAMPLING", "surv_5yrs", "death_5yr", "AGE", "AGE_SAMPLING", "SEX", "OS_STATUS", "OS_MONTHS", "DEATH_DAYS")]
```

```
impute_data$IGHV_MUTATION_STATUS <- as.factor(impute_data$IGHV_MUTATION_STATUS)
impute_data$X11q_status <- as.factor(impute_data$X11q_status)
impute_data$X17p_status <- as.factor(impute_data$X17p_status)
impute_data$X18p_status <- as.factor(impute_data$X18p_status)
impute_data$X19q_status <- as.factor(impute_data$X19q_status)
impute_data$X20p_status <- as.factor(impute_data$X20p_status)
impute_data$X2p_status <- as.factor(impute_data$X2p_status)
impute_data$X4p_status <- as.factor(impute_data$X4p_status)
impute_data$X6q_status <- as.factor(impute_data$X6q_status)
impute_data$X8p_status <- as.factor(impute_data$X8p_status)
impute_data$tri12_status <- as.factor(impute_data$tri12_status)
impute_data$RAI_AT_DIAGNOSIS <- as.factor(impute_data$RAI_AT_DIAGNOSIS)
impute_data$RAI_AT_SAMPLING <- as.factor(impute_data$RAI_AT_SAMPLING)
impute_data$surv_5yrs <- as.numeric(impute_data$surv_5yrs)
impute_data$death_5yr <- as.factor(impute_data$death_5yr)
impute_data$AGE <- as.numeric(impute_data$AGE)
impute_data$AGE_SAMPLING <- as.numeric(impute_data$AGE_SAMPLING)
impute_data$SEX <- as.factor(impute_data$SEX)
impute_data$OS_STATUS <- as.factor(impute_data$OS_STATUS)
impute_data$OS_MONTHS <- as.numeric(impute_data$OS_MONTHS)
impute_data$DEATH_DAYS <- as.numeric(impute_data$DEATH_DAYS)
```

```
imputed_out <- mice(impute_data, m = 5, seed = 500)
```

```
##
## iter imp variable
## 1 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 1 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 1 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 1 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 1 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 2 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 2 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 2 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 2 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 2 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 3 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 3 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
## 3 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_status
```

```
## 3 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 3 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 5 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 5 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 5 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 5 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 5 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
```

```
## Warning: Number of logged events: 1
```

```
data_complete <- complete(imputed_out, 1)
```

```
data$IGHV_MUTATION_STATUS_IMPUTED <- data_complete$IGHV_MUTATION_STATUS
```

```
summary(as.factor(data$IGHV_MUTATION_STATUS_IMPUTED))
```

```
## mutated unmutated
```

```
## 591 557
```

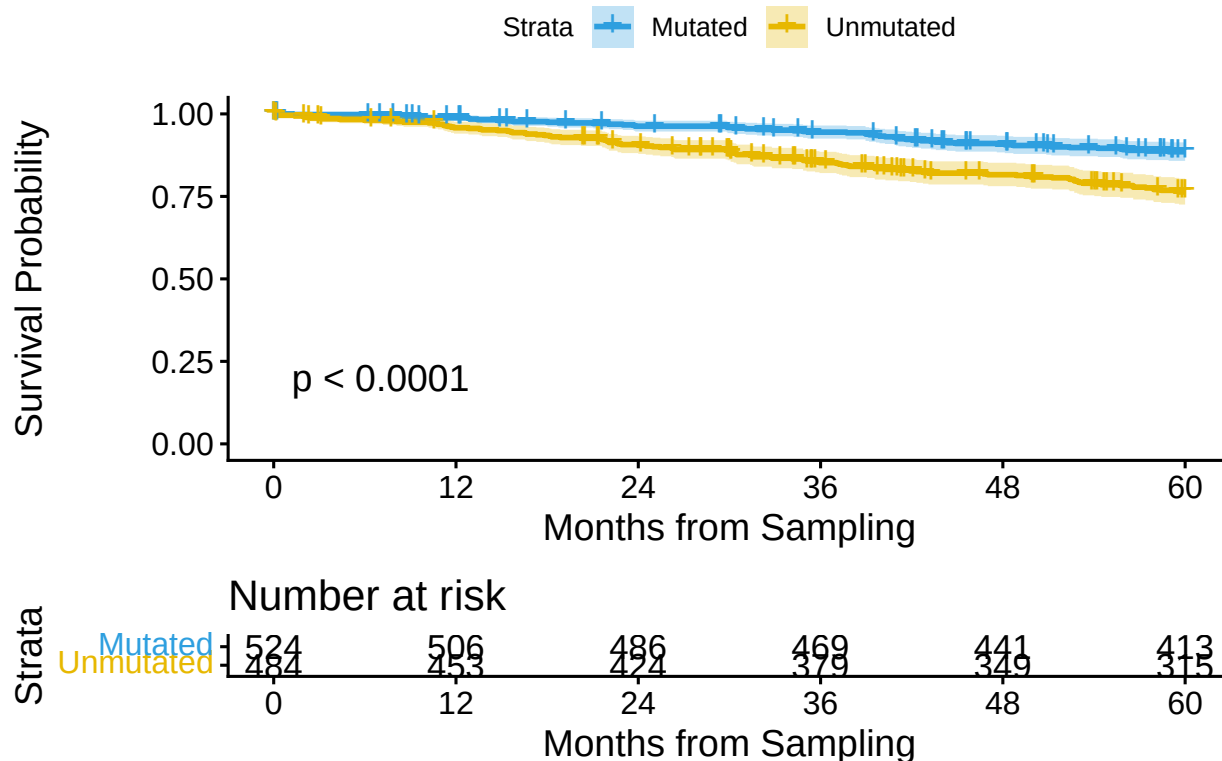
```
fit_5yr <- survfit(Surv(os_months_5yr, death_5yr_status_num) ~ IGHV_MUTATION_STATUS_IMPUTED, data = data)
```

```
ggsurvplot(
  fit_5yr,
  data = data,
  pval = TRUE,
  risk.table = TRUE,
  conf.int = TRUE,
  xlim = c(0, 60),
  break.time.by = 12,
  palette = c("#2E9FDF", "#E7B800"),
  title = "5-Year Overall Survival: IGHV Imputed",
  legend.labs = c("Mutated", "Unmutated"),
  xlab = "Months from Sampling",
  ylab = "Survival Probability"
)
```

```
## Ignoring unknown labels:
```

```
## * colour : "Strata"
```

5-Year Overall Survival: IGHV Imputed



```
cox_ighv <- coxph(Surv(os_months_5yr, death_5yr_status_num) ~ IGHV_MUTATION_STATUS_IMPUTED, data = data,
summary(cox_ighv)
```

```
## Call:
## coxph(formula = Surv(os_months_5yr, death_5yr_status_num) ~ IGHV_MUTATION_STATUS_IMPUTED,
##       data = data)
##
##      n= 1008, number of events= 164
##      (140 observations deleted due to missingness)
##
##               coef exp(coef) se(coef)      z Pr(>|z|)
## IGHV_MUTATION_STATUS_IMPUTEDunmutated 0.8067      2.2405   0.1640 4.917 8.77e-07
##
## IGHV_MUTATION_STATUS_IMPUTEDunmutated ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## IGHV_MUTATION_STATUS_IMPUTEDunmutated      2.24    0.4463    1.624    3.09
##
## Concordance= 0.6 (se = 0.019 )
## Likelihood ratio test= 25.59 on 1 df,  p=4e-07
## Wald test               = 24.18 on 1 df,  p=9e-07
## Score (logrank) test = 25.52 on 1 df,  p=4e-07
```

X17p_status survival

```
summary(as.factor(data$X17p_status))
```

```
##      Loss Unchanged      NA's
##      71      991      86
```

```
impute_data <- data[, c("IGHV_MUTATION_STATUS", "X11q_status", "X17p_status", "X18p_status", "X19q_status",
```

```
impute_data$IGHV_MUTATION_STATUS <- as.factor(impute_data$IGHV_MUTATION_STATUS)
impute_data$X11q_status <- as.factor(impute_data$X11q_status)
impute_data$X17p_status <- as.factor(impute_data$X17p_status)
impute_data$X18p_status <- as.factor(impute_data$X18p_status)
impute_data$X19q_status <- as.factor(impute_data$X19q_status)
impute_data$X20p_status <- as.factor(impute_data$X20p_status)
impute_data$X2p_status <- as.factor(impute_data$X2p_status)
impute_data$X4p_status <- as.factor(impute_data$X4p_status)
impute_data$X6q_status <- as.factor(impute_data$X6q_status)
impute_data$X8p_status <- as.factor(impute_data$X8p_status)
impute_data$tri12_status <- as.factor(impute_data$tri12_status)
impute_data$RAI_AT_DIAGNOSIS <- as.factor(impute_data$RAI_AT_DIAGNOSIS)
impute_data$RAI_AT_SAMPLING <- as.factor(impute_data$RAI_AT_SAMPLING)
impute_data$surv_5yrs <- as.numeric(impute_data$surv_5yrs)
impute_data$death_5yr <- as.factor(impute_data$death_5yr)
impute_data$AGE <- as.numeric(impute_data$AGE)
impute_data$AGE_SAMPLING <- as.numeric(impute_data$AGE_SAMPLING)
impute_data$SEX <- as.factor(impute_data$SEX)
impute_data$OS_STATUS <- as.factor(impute_data$OS_STATUS)
impute_data$OS_MONTHS <- as.numeric(impute_data$OS_MONTHS)
impute_data$DEATH_DAYS <- as.numeric(impute_data$DEATH_DAYS)
```

```
imputed_out <- mice(impute_data, m = 5, seed = 500)
```

```
##
## iter imp variable
## 1 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 1 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 1 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 1 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 1 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 2 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 2 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 2 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 2 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 2 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 3 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 3 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 3 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 3 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 3 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 4 5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
## 5 1 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
```

```
##    5    2 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
##    5    3 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
##    5    4 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
##    5    5 IGHV_MUTATION_STATUS X11q_status X17p_status X18p_status X19q_status X20p_status X2p_
```

```
## Warning: Number of logged events: 1
```

```
data_complete <- complete(imputed_out, 3)
```

```
data$X17p_status_complete <- data_complete$X17p_status
```

```
summary(as.factor(data$X17p_status_complete))
```

```
##      Loss Unchanged
```

```
##      82      1066
```

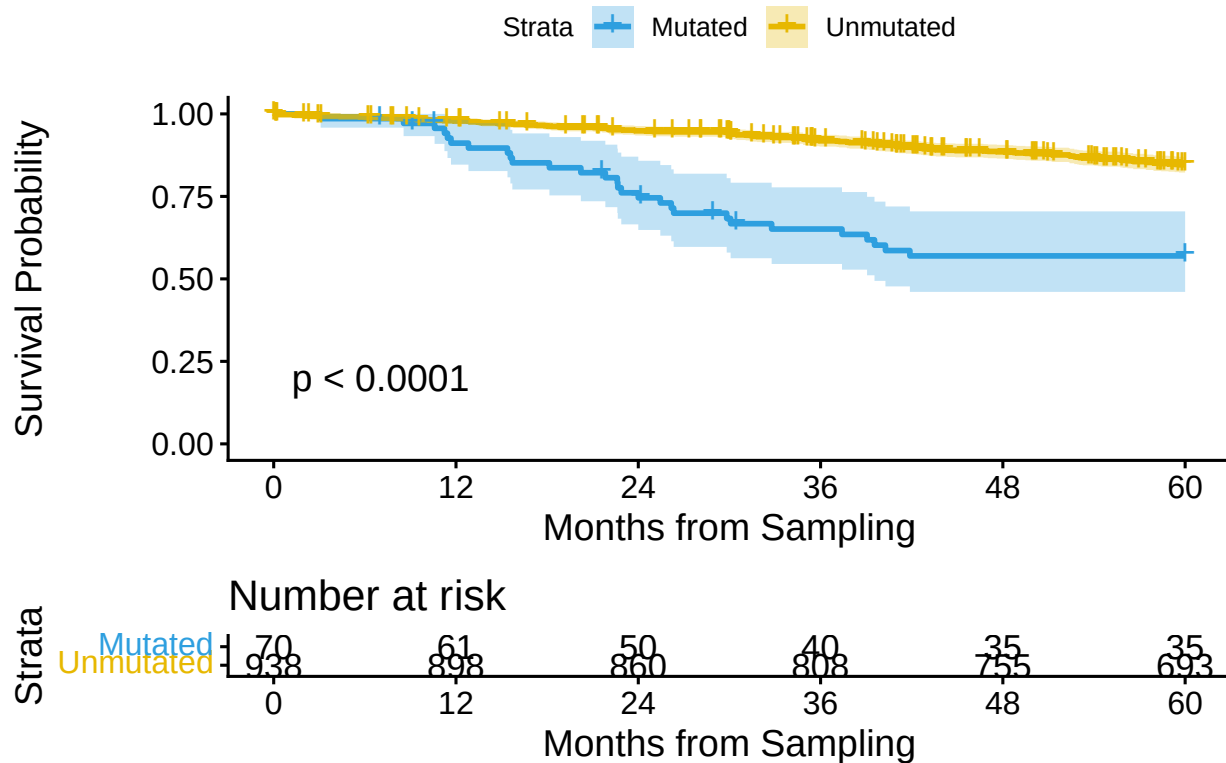
```
fit_5yr <- survfit(Surv(os_months_5yr, death_5yr_status_num) ~ X17p_status_complete, data = data)
```

```
ggsurvplot(
  fit_5yr,
  data = data,
  pval = TRUE,
  risk.table = TRUE,
  conf.int = TRUE,
  xlim = c(0, 60),
  break.time.by = 12,
  palette = c("#2E9FDF", "#E7B800"),
  title = "5-Year Overall Survival: X17p_status",
  legend.labs = c("Mutated", "Unmutated"),
  xlab = "Months from Sampling",
  ylab = "Survival Probability"
)
```

```
## Ignoring unknown labels:
```

```
## * colour : "Strata"
```

5-Year Overall Survival: X17p_status



```
cox_x17p <- coxph(Surv(os_months_5yr, death_5yr_status_num) ~ X17p_status_complete, data = data)
summary(cox_x17p)
```

```
## Call:
## coxph(formula = Surv(os_months_5yr, death_5yr_status_num) ~ X17p_status_complete,
##       data = data)
##
## n= 1008, number of events= 164
## (140 observations deleted due to missingness)
##
##               coef exp(coef) se(coef)      z Pr(>|z|)
## X17p_status_completeUnchanged -1.3074    0.2705    0.2080 -6.286 3.26e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##               exp(coef) exp(-coef) lower .95 upper .95
## X17p_status_completeUnchanged    0.2705      3.697    0.1799    0.4067
##
## Concordance= 0.563 (se = 0.015 )
## Likelihood ratio test= 29.51 on 1 df,  p=6e-08
## Wald test               = 39.51 on 1 df,  p=3e-10
## Score (logrank) test = 45.45 on 1 df,  p=2e-11
```

RAI_AT_DIAGNOSIS

```
summary(as.factor(data$RAI_AT_DIAGNOSIS))
```

```
##    0    1    2    3    4 NA's
```

```
## 819 82 78 9 20 140
```

```
impute_data <- data[, c("RAI_AT_DIAGNOSIS", "IGHV_MUTATION_STATUS", "X11q_status", "X17p_status", "X18p_s
```

```
impute_data$IGHV_MUTATION_STATUS <- as.factor(impute_data$IGHV_MUTATION_STATUS)
impute_data$X11q_status <- as.factor(impute_data$X11q_status)
impute_data$X17p_status <- as.factor(impute_data$X17p_status)
impute_data$X18p_status <- as.factor(impute_data$X18p_status)
impute_data$X19q_status <- as.factor(impute_data$X19q_status)
impute_data$X20p_status <- as.factor(impute_data$X20p_status)
impute_data$X2p_status <- as.factor(impute_data$X2p_status)
impute_data$X4p_status <- as.factor(impute_data$X4p_status)
impute_data$X6q_status <- as.factor(impute_data$X6q_status)
impute_data$X8p_status <- as.factor(impute_data$X8p_status)
impute_data$tri12_status <- as.factor(impute_data$tri12_status)
impute_data$RAI_AT_DIAGNOSIS <- as.factor(impute_data$RAI_AT_DIAGNOSIS)
impute_data$RAI_AT_SAMPLING <- as.factor(impute_data$RAI_AT_SAMPLING)
impute_data$surv_5yrs <- as.numeric(impute_data$surv_5yrs)
impute_data$death_5yr <- as.factor(impute_data$death_5yr)
impute_data$AGE <- as.numeric(impute_data$AGE)
impute_data$AGE_SAMPLING <- as.numeric(impute_data$AGE_SAMPLING)
impute_data$SEX <- as.factor(impute_data$SEX)
impute_data$OS_STATUS <- as.factor(impute_data$OS_STATUS)
impute_data$OS_MONTHS <- as.numeric(impute_data$OS_MONTHS)
impute_data$DEATH_DAYS <- as.numeric(impute_data$DEATH_DAYS)
```

```
imputed_out <- mice(impute_data, m = 5, maxit = 5, seed = 500)
```

```
##
```

```
## iter imp variable
```

##	1	2	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	1	2	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	1	3	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	1	4	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	1	5	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	2	1	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	2	2	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	2	3	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	2	4	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	2	5	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	3	1	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	3	2	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	3	3	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	3	4	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	3	5	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	4	1	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	4	2	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	4	3	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	4	4	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	4	5	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	5	1	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	5	2	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	5	3	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	5	4	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status
##	5	5	RAI_AT_DIAGNOSIS	IGHV_MUTATION_STATUS	X11q_status	X17p_status	X18p_status	X19q_status


```

## Warning: Number of logged events: 1
data_complete <- complete(imputed_out, 1)

data$RAI_AT_DIAGNOSIS_COMPLETE <- data_complete$RAI_AT_DIAGNOSIS

summary(as.factor(data_complete$RAI_AT_DIAGNOSIS))

##    0    1    2    3    4
## 922 101  92  10  23

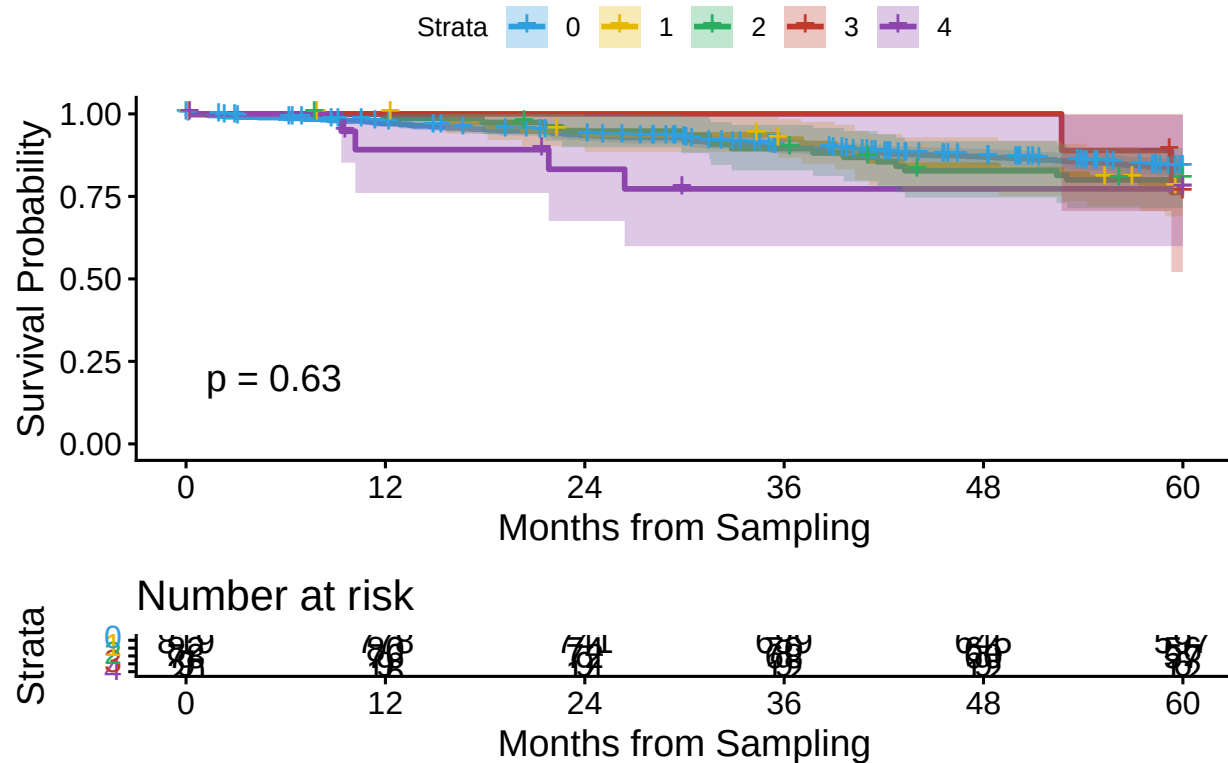
fit_5yr <- survfit(Surv(os_months_5yr, death_5yr_status_num) ~ RAI_AT_DIAGNOSIS_COMPLETE, data = data)

ggsurvplot(
  fit_5yr,
  data = data,
  pval = TRUE,
  risk.table = TRUE,
  conf.int = TRUE,
  xlim = c(0, 60),
  break.time.by = 12,
  palette = c("#2E9FDF", "#E7B800", "#27ae60", "#c0392b", "#8e44ad"),
  title = "5-Year Overall Survival: RAI_AT_DIAGNOSIS",
  legend.labs = c("0", "1", "2", "3", "4"),
  xlab = "Months from Sampling",
  ylab = "Survival Probability"
)

## Ignoring unknown labels:
## * colour : "Strata"

```

5-Year Overall Survival: RAI_AT_DIAGNOSIS



```
cox_rai <- coxph(Surv(os_months_5yr, death_5yr_status_num) ~ RAI_AT_DIAGNOSIS, data = data)
summary(cox_rai)
```

```
## Call:
## coxph(formula = Surv(os_months_5yr, death_5yr_status_num) ~ RAI_AT_DIAGNOSIS,
##       data = data)
##
## n= 1008, number of events= 164
## (140 observations deleted due to missingness)
##
##               coef exp(coef) se(coef)      z Pr(>|z|)
## RAI_AT_DIAGNOSIS 0.12246   1.13027  0.08648  1.416   0.157
##
##               exp(coef) exp(-coef) lower .95 upper .95
## RAI_AT_DIAGNOSIS      1.13    0.8847   0.954   1.339
##
## Concordance= 0.521 (se = 0.016 )
## Likelihood ratio test= 1.84 on 1 df,  p=0.2
## Wald test               = 2.01 on 1 df,  p=0.2
## Score (logrank) test = 2.02 on 1 df,  p=0.2
```